

A Clinical Workflow Integration Layer for Tuberculosis Diagnosis in Chest X-ray Imaging

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Abstract—Tuberculosis (TB) remains a leading infectious cause of death worldwide, with the Philippines ranking among the highest-burden countries globally. Computer-aided detection (CAD) for TB on chest X-ray has reached sufficient maturity that the World Health Organization (WHO) recommended its use for screening in 2021, and six commercial products met WHO performance standards in a 2025 policy update. Despite strong detection accuracy, the transition from laboratory performance to reliable hospital deployment remains incomplete: workflow integration, input quality assurance, clinician-facing explanation of findings, and post-deployment monitoring are underdeveloped in current systems. This paper presents a prototype clinical workflow integration layer designed to wrap around a mature TB CAD backbone. The lead contribution is a clinically grounded multimodal explanation module that translates model findings into natural-language statements linking suspected abnormalities to their visual basis on the radiograph, a feature not clearly present as a deployed capability in the commercial products reviewed. Supporting modules include an input quality gate, a structured evidence interface, and a post-deployment monitoring dashboard. The system is framed as an academic prototype for hospital chest X-ray screening workflows, with preliminary evaluation focused on feasibility and clinical utility rather than claims of algorithmic novelty in the underlying classifier.

Keywords—Tuberculosis; Computer-Aided Detection; Chest X-Ray; Clinical Workflow; Medical Imaging; Computer Vision

I. INTRODUCTION

Tuberculosis (TB) is an airborne bacterial infection caused by *Mycobacterium tuberculosis* that primarily affects the lungs. It spreads when an infected person coughs, sneezes, or speaks, releasing bacteria that others may inhale. TB is both preventable and curable, yet it remains one of the leading infectious causes of death worldwide. The World Health Organization (WHO) estimated that 10.8 million people fell ill with TB in 2023, and approximately 1.25 million died from the disease [1]. Low- and middle-income countries bear a disproportionate share of this burden.

The Philippines is among the countries most severely affected. WHO classifies it as a high-burden country for TB, ranking fourth globally in estimated incidence [1]. In 2023, the country reported an estimated incidence rate of 638 per 100,000 population, one of the highest in the world [2]. TB accounts for a substantial share of infectious disease mortality in the Philippines, compounded by limited access to diagnostics in rural areas, delayed presentation, and an overburdened public health infrastructure.

Chest X-ray (CXR) is the most widely used imaging modality for TB screening. A standard posteroanterior chest

radiograph can reveal patterns associated with pulmonary TB, including upper-lobe infiltrates, cavitary lesions, pleural effusions, and lymphadenopathy. In screening settings, CXR serves as a triage tool: patients whose radiographs show findings suggestive of TB are referred for confirmatory bacteriological testing, typically sputum smear microscopy or molecular assays such as GeneXpert. This two-stage workflow, radiographic triage followed by laboratory confirmation, is central to national TB control programs in high-burden countries [3].

Interpreting chest radiographs for TB depends on trained readers. In many high-burden countries, radiologists are scarce, particularly in primary care facilities and rural health units where screening volumes are highest [3]. This bottleneck has driven the development of computer-aided detection (CAD) systems that use artificial intelligence (AI) to analyze CXR images and flag those suggestive of TB.

A. Maturity of TB CAD Systems

CAD for TB on chest X-ray is no longer an emerging technology. In 2021, WHO issued a policy recommendation supporting CAD as an alternative to human readers for TB screening and triage [4]. In June 2025, WHO reported that six commercial CAD products met WHO target performance standards for TB screening in individuals aged 15 years and older [5]. The WHO target product profile specifies minimum sensitivity of 90% and specificity of 70% for triage, benchmarks that multiple products now meet or exceed [6].

The broader evidence base reinforces this maturity. A systematic review of machine learning for chest X-ray interpretation found strong diagnostic performance across a range of thoracic conditions, though it also noted limited prospective validation and real-world deployment evidence [7]. A meta-analysis of CAD accuracy during active case finding for pulmonary TB in Africa reported pooled sensitivity of 0.87 and specificity of 0.74, approaching, though not uniformly meeting, the WHO screening target [36]. A 2025 meta-analysis of five commercial CAD products found individual sensitivities ranging from 86% to 91%, with some products meeting the WHO threshold and others falling slightly below depending on the operating point selected [37]. A 2024 reader study demonstrated that AI assistance raised physician diagnostic performance, measured by the area under the receiver operating characteristic curve (a summary statistic capturing accuracy across all decision thresholds), from 0.773 to 0.874 and helped non-radiologist physicians perform on par with radiologists [8]. In a real-world observational study, radiologists showed complete agreement with AI findings in 86.5% of cases, with only 0.5% of findings classified as missed by the AI [9].

These results indicate that the field has moved beyond asking whether AI can achieve useful diagnostic accuracy on chest X-rays. The question now is whether these systems can function reliably and safely when integrated into routine hospital workflows.

B. Related Work

Beyond deployed commercial systems, a substantial body of academic research has produced TB CXR analysis methods and prototypes. Early deep-learning work showed that convolutional neural networks could classify pulmonary TB on chest radiographs at accuracy comparable to radiologists [15], and subsequent work produced efficient architectures pairing fast screening with saliency-based visualization [16]. The TBX11K benchmark reframed TB CAD as a detection problem with region-level annotations and released reference detectors [45], [46], which the present system adopts as its backbone. On the explanation side, research prototypes have begun coupling detectors or image encoders with language models: Danu et al. at Siemens Healthineers couple a thoracic abnormality detector with a domain-adapted large language model to generate chest-X-ray findings [13], while general medical vision-language models such as LLaVA-Med [14] and CXR-LLaVA [27] demonstrate image-conditioned radiology text generation, and recent surveys catalogue the rapid growth of vision-language models for medical report generation and visual question answering [28]. These remain research prototypes rather than deployed clinical features: trained weights are frequently unavailable, evaluation is typically retrospective, and image-conditioned generative models in particular are prone to hallucination [13], a risk that motivates the grounding-by-construction design adopted here.

Among deployed systems, a feature comparison across five commercial CAD products, qXR (Qure.ai), CAD4TB+ (Delft Imaging), Lunit INSIGHT CXR, Annalise Enterprise CXR, and Siemens AI-Rad Companion, summarized in Table I, shows that most provide TB detection, heatmap-based localization, and in some cases worklist prioritization and analytics. Based on official product documentation, two capabilities were not clearly present as deployed features:

- **Natural-language explanation of findings.** No reviewed product provides a deployed feature explaining diagnostic conclusions in clinician-facing causal language. As discussed above, the most directly relevant approach, Danu et al. at Siemens Healthineers [13], remains a research prototype not exposed as a documented feature in Siemens' commercial AI-Rad Companion product [23].
- **Integrated workflow.** No reviewed system combines input quality gating and natural-language explanation into a single layer.

The proposed system occupies this gap as a prototype demonstrating how such features could be integrated.

C. Gaps in Clinical Workflow Integration

Despite strong detection performance, reviews of CXR AI deployment consistently identify a gap between laboratory accuracy and clinical implementation. A review of practical

AI in chest radiography found that only a small proportion of published studies are prospective or clinically embedded, and that workflow integration, local validation, and human factors remain under-studied [10]. The same body of literature argues that clinical expertise, usability, and implementation context remain essential even when model performance is strong [7], [10]. Real-world deployment studies reinforce this: an implementation of qXR for TB screening in India identified barriers including software installation, workflow integration, network connectivity, and limited human resources for interpreting findings [33]; a deployment study in Vietnam found only 47.3% concordance between physician and CAD interpretations at three months [34]; and early user interviews across multiple countries found that clinicians valued CAD most when it complemented rather than replaced clinical judgment [35]. Usability testing of a CXR AI tool in a simulated clinical workflow reported that users wanted clearer explanatory support and suggested the tool would be most useful in understaffed radiology departments, for trainees, and for non-radiologist physicians [11].

Several specific gaps emerge from this literature:

- **Input quality.** CAD systems are trained on curated datasets. In routine hospital use, images may be poorly positioned, clipped, rotated, underexposed, or photographed from analog film. Poor input quality can degrade performance, but most systems do not expose a user-facing quality assessment before inference [10], [12].
- **Explanation of findings.** Current commercial CAD products provide heatmap overlays showing image regions that influenced the prediction. A systematic review of explainability methods in medical imaging AI found that saliency-based approaches often show low overlap with expert-identified regions of interest and do not convey the contextual reasoning clinicians use during diagnosis [25]. Alternative explanation approaches, including textual descriptions and case-based reasoning, have been proposed as more clinically meaningful [26]. Non-radiologists may benefit from natural-language explanations linking a conclusion to its visual evidence [13].
- **Post-deployment monitoring.** A deployed system's performance may change due to population shifts, equipment changes, or software updates. The U.S. Food and Drug Administration (FDA) has identified data drift, concept drift, and model drift as risks to AI-enabled medical devices and has initiated research programs on proactive post-market monitoring [29]. Reviews note the need for ongoing monitoring, but this remains an operational gap in practice [10].

D. Scope and Contribution

This paper addresses the workflow integration gap by presenting a prototype **clinical workflow integration layer** for TB CAD in hospital CXR screening. The system wraps clinician-facing modules around a classification backbone:

- 1) A **multimodal explanation module** that generates natural-language descriptions of AI-detected findings,

TABLE I. FEATURE PRESENCE ACROSS REVIEWED COMMERCIAL TB CAD PRODUCTS AND THE PROPOSED CLINICAL WORKFLOW INTEGRATION LAYER, BASED ON OFFICIAL PRODUCT DOCUMENTATION FOR QXR [19], CAD4TB+ [20], LUNIT INSIGHT CXR [21], ANNALISE ENTERPRISE CXR [22], AND SIEMENS AI-RAD COMPANION [23]. A CHECKMARK INDICATES A DOCUMENTED FEATURE; A BLANK INDICATES ABSENCE OR LACK OF CLEAR EVIDENCE IN THE REVIEWED MATERIALS.

Feature	qXR	CAD4TB+	Lunit	Annalise	Siemens	Proposed
TB-specific support	✓	✓	✓			✓
Heatmap / overlay localization	✓	✓	✓	✓	✓	✓
Worklist / prioritization	✓		✓	✓		
Configurable threshold tuning		✓				✓
Monitoring / analytics dashboard	✓	✓	✓			✓
User-facing input quality gate					✓	✓
Natural-language finding explanation						✓

linking suspected abnormalities to localized visual evidence. This is the lead contribution.

- 2) An **input quality gate** that screens radiographs for acquisition defects before inference.
- 3) A **structured evidence interface** presenting the TB score, risk classification, quality status, and localized evidence in a compact clinical panel.
- 4) A **post-deployment monitoring dashboard** tracking system behavior over time.

The primary claim is one of clinical utility and feasibility, that this combination addresses documented needs in hospital TB screening, not algorithmic novelty in the underlying detection model.

The system addresses four design goals, each corresponding to a documented failure mode or usability gap:

- 1) **Reject or flag unsafe inputs before inference.** Prevent the classifier from operating on images likely to produce unreliable predictions.
- 2) **Provide clinically grounded explanations.** Translate model evidence into natural-language statements describing what was found, where, and why it matters clinically.
- 3) **Present evidence in a workflow-friendly format.** Consolidate risk score, quality status, heatmap, and explanation into a single panel for rapid triage review.
- 4) **Monitor deployment quality.** Track operational metrics to detect degradation or drift after hospital deployment.

II. MATERIALS AND METHODS

A. System Architecture

The system processes a chest X-ray through a sequential pipeline:

- 1) **Image ingestion.** A digital chest radiograph (DICOM or standard image format) is received from the hospital imaging system or uploaded directly.
- 2) **Input quality assessment.** The image is evaluated for acquisition quality before diagnostic inference.
- 3) **TB classification.** If the image passes quality screening, it is processed by the TB scoring module, producing a continuous probability score and spatial activation maps.

- 4) **Multimodal explanation generation.** For classified cases, the system generates a natural-language explanation linking the predicted finding to localized visual evidence.
- 5) **Clinical interface rendering.** All outputs are presented in a structured evidence panel.
- 6) **Monitoring and logging.** Each case and its metadata are logged for the monitoring dashboard.

Beyond the linear pipeline, two cross-cutting components operate within the architecture.

Structured evidence interface. All outputs are consolidated into a single panel for rapid clinical review: the TB probability score as a numeric value and color-coded risk band (low, moderate, high); the quality status with warnings for detected issues; a heatmap overlay showing regions influencing the prediction; and the natural-language explanation from the multimodal module. The panel uses a traffic-light color scheme for risk and quality status and is designed to be reviewable in under 30 seconds during routine triage.

Post-deployment monitoring. A monitoring dashboard tracks operational metrics to detect behavioral changes after deployment, drawing on the literature on post-deployment medical-AI monitoring [50], [51], [52], [30] and metrics published by existing TB CAD products [19], [20], [21]. Metrics are organized into five groups: input-side (case volume, quality-gate rejection rate, scanner/site/demographic mix relative to training); prediction-side (TB-score distribution, flag rate, per-finding trajectory, calibration metrics); outcome-side (radiologist override rate, CAD-human discordance, and bacteriologically confirmed TB yield among CAD-positive cases, explicitly called for in WHO guidance [3]); drift (Population Stability Index and Kolmogorov-Smirnov statistics on score distributions, with PSI < 0.1 stable, 0.1–0.25 flagged, ≥ 0.25 a major shift [51], plus higher-dimensional embedding drift detection [52], [53]); and fairness (sensitivity, specificity, and positive predictive value stratified by clinically relevant subgroups with per-subgroup confidence intervals, as required by regulatory guidance [29]). The dashboard does not autonomously adjust system behavior; it enables human intervention when metrics indicate degradation or drift [10], [30].

B. Input Quality Assessment

The input quality gate evaluates each radiograph before it reaches the classifier. It assigns one of four status labels:

- **Acceptable:** meets acquisition criteria; proceeds to classification.
- **Borderline:** minor quality issues (slight rotation, sub-optimal exposure) that may reduce confidence. Classification proceeds with a quality warning attached.
- **Repeat recommended:** significant defects (clipped lung fields, severe underexposure) likely to compromise reliability. The system recommends re-acquisition.
- **Manual review required:** non-standard input (photographed analog film, lateral view, non-chest image). The case is routed to human review.

The gate is a hybrid pipeline combining DICOM header checks, a learned view classifier, and an anatomical-segmentation-based sanity check. Concretely: (i) DICOM tags (`ViewPosition`, `BodyPartExamined`, `PhotometricInterpretation`) are read first; any study with a non-frontal `ViewPosition` is routed to manual review, and inconsistent `BodyPartExamined` triggers a rejection. (ii) For cases lacking reliable DICOM headers, a small view classifier trained on frontal / lateral labels from PadChest [49] serves as a fallback. (iii) The image is passed through the pretrained anatomical segmentation network from the `torchxrayvision` library [48], which produces per-pixel masks for left lung, right lung, heart, and other thoracic structures. The gate computes three summary statistics from these masks: the lung-area-to-image-area ratio (a field-of-view score, flagging clipped fields), the lateral offset of the lung centroid relative to image center (a rotation proxy), and the presence of both lung masks above a minimum area (a non-chest-image check). (iv) Classical image statistics, Laplacian variance for blur and 1st/99th percentile intensity gaps for under- or over-exposure, are thresholded on development data and added to the rule set. The combined rules map each image to one of the four status labels above.

This design is motivated by evidence that CXR AI performance degrades when applied to images differing from the training distribution. A systematic review of cross-population domain shift found that 81% of deep learning algorithms in radiology exhibited decreased accuracy on external datasets [38], and shortcut learning, where models rely on acquisition artifacts rather than pathological features, has been documented as a failure mode in CXR AI [12], [10]. The gate is engineered rather than novel; it is cited as prior art that a clinically adequate CXR screening system has a well-defined acquisition-quality model upstream of inference.

Local threshold calibration matters for adapting CAD to site-specific prevalence and clinical context. WHO has published calibration guidance [17], and CAD4TB+ offers dynamic threshold adjustment [20]. A 2025 study of qXR in the Philippines demonstrated the sensitivity–specificity tradeoff from threshold selection [18]. The classifier operating point is therefore retained as a configurable parameter, but not positioned as the headline contribution, it addresses a partially solved problem with existing guidance.

C. Tuberculosis Detection

The tuberculosis detection module is the diagnostic backbone. It is not a contribution of this paper; it is used as an established component consistent with WHO’s finding that multiple TB CAD products already meet screening performance standards [5].

1) *Dataset:* The detector is trained on **TBX11K**, a public dataset of 11,200 chest radiographs with expert-annotated bounding boxes for active-TB and latent-TB regions [45]. TBX11K was selected because (i) it is paired with reference detector baselines in the same publication, so reported detection metrics transfer cleanly; (ii) its canonical 6,600 / 1,800 / 2,800 training, validation, and test split supports reproducible evaluation; and (iii) it is the largest publicly available TB-specific CXR dataset with region-level annotations.

2) *Data Preprocessing:* Images and annotations follow the canonical TBX11K training, validation, and test split so that detection metrics are directly comparable to the published baselines. Because the TBX11K annotation schema labels regions as *active_tb* or *latent_tb* rather than as specific findings (e.g., “cavitation”), each detected region is, in a subsequent pass, intersected with a pretrained anatomical lung-zone segmentation [48] to obtain an anatomical descriptor (lung side and zone); this descriptor is consumed by the multimodal explanation module described in Section II-D.

3) *Model Architecture:* The module is implemented as an object-detection network that produces bounding boxes localizing TB-related regions on a chest radiograph, alongside per-region probabilities and an image-level TB score. Object detection is a form of deep learning in which the model learns to both classify and spatially localize visual findings, in this case, regions suggestive of active or latent tuberculosis. We adopt the detector baselines released with the TBX11K benchmark [45], [46], specifically a Faster R-CNN with a ResNet-50 backbone as the primary architecture, with the transformer-based SymFormer variant [46] retained as a stretch baseline. Faster R-CNN was chosen for feasibility: it is well-documented, has reference training configurations released in the associated public repository, and its detection output format (axis-aligned bounding boxes with class probabilities) is directly compatible with the downstream explanation module. The final per-image output consists of (1) an image-level TB probability score, (2) a list of detected regions with bounding-box coordinates, TB-type class probability, anatomical zone, and candidate finding label, and (3) a spatial activation map retained for the heatmap overlay.

4) *Model Training:* The detector is trained using the reference training configurations released with the TBX11K baselines [45], [46].

D. Multimodal Explanation Module

The multimodal explanation module is the lead contribution. It translates the classifier’s evidence into a natural-language statement that a clinician can read, understand, and act on without interpreting heatmaps or understanding the model’s internals. Evidence for the value of natural-language explanation in CXR AI comes from user studies and clinician-interview research, though the demand picture is mixed. In a

user-centered evaluation of explainability techniques on chest radiographs with 26 clinicians at a National Health Service trust in the United Kingdom, a majority stated that descriptive sentences should be added alongside visual explanations, and roughly half reported that colored heatmap overlays hindered readability [39]. A multisite prospective study of 220 physicians reading chest radiographs found that the form of AI explanation materially affected diagnostic performance and physician trust, indicating that explanation modality, not only the presence of an explanation, shapes clinical use [40]. Broader interview-based studies of AI decision support in medicine have surfaced consistent requests for contextual reasoning and explanatory information rather than predictions alone [41], [42]. In the TB CAD setting specifically, a qualitative study of radiologists using a commercial TB CXR product found opinion divided: one participant reported struggling to understand why the system reached a given conclusion, while another characterized the tool as an engine whose internal workings were not clinically relevant to them [43]. The module is therefore positioned as most useful for a radiologist-in-the-loop or confirmatory reader reviewing flagged cases, rather than as a universally demanded feature for frontline screeners who primarily act on the abnormality score.

Given a radiograph and the classifier's output, the module generates a statement identifying the suspected finding (e.g., cavitary lesion, upper-lobe infiltrate, pleural effusion), its anatomical location (e.g., right upper lobe, left costophrenic angle), and a clinical reasoning link connecting the finding to the TB classification. An example output: "*TB suspected. The model identified an area of increased opacity in the right upper lobe consistent with an infiltrate. Upper-lobe infiltrates are among the most common radiographic findings in active pulmonary tuberculosis.*" This differs from heatmap-only explanations in two ways. First, it names the finding in clinical vocabulary rather than highlighting a region. Second, it provides a reason, grounded in radiographic knowledge, for why that finding supports a TB classification.

1) *Dataset*: For finer-grained finding labels, such as cavity, consolidation, infiltrate, and pleural effusion, the module uses the Shenzhen TB annotation set [47], which supplies pixel-level masks for 19 abnormality types on 336 TB-positive cases.

2) *Data Preprocessing*: Each detected region's bounding-box centroid is intersected with the anatomical lung-zone segmentation from the quality-assessment stage [48] to produce an anatomical descriptor (left or right; upper, middle, or lower zone). Each region is then mapped to a candidate finding label drawn from the Shenzhen TB abnormality taxonomy [47]; when the TB detector flags only *active_tb* or *latent_tb* without a specific finding type, a secondary finding classifier trained on the Shenzhen annotations produces a label.

3) *Model Architecture*: The module operates as a post-processing stage after the TB detection module has produced bounding boxes and class probabilities. It is deliberately structured as a two-stage pipeline, *structured templating* followed by *large language model (LLM) smoothing*, in which the image itself is withheld from the LLM. This architecture is chosen to prevent the LLM from inventing findings the detector did not produce: because the LLM only receives a structured record of detected regions, it cannot narrate anatomical findings outside that record.

The most closely related published system is the two-stage pipeline of Danu et al. [13], in which a multi-class thoracic detector emits class probabilities that are concatenated as a prompt to a domain-adapted LLM (RadBloomz) for report generation. The present architecture takes the same decomposition as prior art but differs in two respects: (1) it restricts the controlled vocabulary to TB-relevant findings rather than a general thoracic taxonomy, and (2) it does not rely on a fine-tuned biomedical LLM, whose weights are not publicly available in the Danu et al. work; instead it uses a generic hosted LLM constrained by prompt and template. The authors of the prior work themselves report that their end-to-end approach suffers from hallucination and repetition in some outputs [13], which motivates the grounding-by-construction approach here.

Consuming the (finding, anatomical zone, probability) records produced in the preprocessing stage, the generation pipeline proceeds as follows:

- 1) **Templated sentence generation.** Each (finding, anatomical zone, probability) triple is slot-filled into a controlled sentence template, producing a deterministic structured record in the form "*{finding} in {zone} (prob:.0% confidence).*"
- 2) **Natural-language smoothing.** The structured list of template sentences is passed, without the radiograph image, to a hosted general-purpose LLM with a system prompt that instructs it to rewrite the list as a single fluent clinical paragraph, add no findings absent from the list, and preserve anatomical and probability information verbatim.
- 3) **Post-hoc verification.** Output text is parsed with a named-entity recognition pass to check that every finding mentioned in the output appears in the input list. Outputs with unsupported entities are rejected and replaced by the deterministic templated text alone.

4) *Model Training*: The secondary finding classifier is trained on the Shenzhen TB annotations [47]; the natural-language smoothing stage uses a hosted general-purpose LLM without task-specific fine-tuning, constrained by prompt and template.

III. RESULTS AND DISCUSSION

A. Input Quality Assessment

Evaluation of the input quality assessment module is in progress.

B. Tuberculosis Detection

Detection results on the TBX11K test split are pending.

C. Multimodal Explanation Model

Qualitative and quantitative evaluation of generated explanations is pending.

D. Future Work

- **Prospective clinical evaluation.** Hospital deployment with measurement of diagnostic outcomes, clinician behavior, and patient impact.
- **Cost-effectiveness analysis.** Assessment of whether the clinical workflow integration layer reduces unnecessary testing, missed cases, or clinical time. Cost data for AI-CAD workflows in high-burden settings are fragmented; a local pilot study would be needed to produce credible estimates.
- **Photographed-film workflows.** Extension of the quality gate to handle smartphone-captured analog radiographs, common in rural health facilities in high-burden countries, with dedicated evaluation of domain shift [24].
- **Formal explanation validation.** Systematic comparison of generated explanations against radiologist reports.
- **Trainee-oriented evaluation.** Primary user research with residents and non-radiologist physicians to determine whether a dedicated trainee mode is actually demanded, what form it should take, and whether it produces durable skill gains rather than in-session dependence.

E. Limitations

This work has several limitations. First, the system has not been evaluated in a live clinical environment; evaluation relies on retrospective data and simulated workflows. Second, the explanation module's outputs have not been validated against expert radiologist interpretations at scale; incorrect explanations could reduce rather than build trust. Third, quality gate thresholds were set on development data and may require recalibration for different imaging environments. Fourth, the TB detection module processes only a single frontal (posteroanterior or anteroposterior) chest radiograph per case, a constraint shared by all WHO-approved TB CAD products, which are trained exclusively on frontal projections. Unlike a clinical PACS hanging protocol that presents current and prior images side by side, the system cannot compare serial radiographs. This restricts its use to screening of new patients at a single time point; it cannot track radiographic change over the course of TB treatment or detect progression and regression across follow-up examinations. Finally, the multimodal explanation module is a prototype, not a clinically validated explanation system: the grounding-by-construction approach reduces but does not eliminate error, since templated sentences can still propagate incorrect detector outputs and the LLM smoothing step could paraphrase findings in ways that shift clinical meaning, which is why the post-hoc verification step is included.

IV. CONCLUSION

DECLARATION ON GENERATIVE AI

Generative AI tools were used to assist in drafting and refining language during manuscript preparation. The conception of the research, development of methodology, and all conclusions are the work of the authors.

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REFERENCES

- [1] World Health Organization, "Global Tuberculosis Report 2024," Geneva: WHO, 2024. [Online]. Available: <https://www.who.int/teams/global-tuberculosis-programme/tb-reports/global-tuberculosis-report-2024>
- [2] World Health Organization, "Tuberculosis profile: Philippines," 2024. [Online]. Available: https://worldhealthorg.shinyapps.io/tb_profiles/
- [3] World Health Organization, "WHO consolidated guidelines on tuberculosis. Module 2: Screening – Systematic screening for tuberculosis disease," Geneva: WHO, 2021. [Online]. Available: <https://www.who.int/publications/i/item/9789240022676>
- [4] World Health Organization, "WHO operational handbook on tuberculosis. Module 2: Screening," Geneva: WHO, 2021. [Online]. Available: <https://www.who.int/publications/i/item/9789240110373>
- [5] World Health Organization, "WHO approves six software products for computer-aided detection of TB on chest X-ray," June 2025. [Online]. Available: <https://www.who.int/news/item/11-06-2025-who-approves-six-software-products-for-computer-aided-detection-of-tb-on-chest-x-ray>
- [6] World Health Organization, "High-priority target product profiles for new tuberculosis diagnostics: report of a consensus meeting," Geneva: WHO, 2014. [Online]. Available: <https://www.who.int/publications/i/item/WHO-HTM-TB-2014.18>
- [7] H. K. Ahmad *et al.*, "Machine learning augmented interpretation of chest X-rays: A systematic review," *Diagnostics*, vol. 13, no. 4, art. 743, 2023. [Online]. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC9955112/>
- [8] P. G. Anderson *et al.*, "Deep learning improves physician accuracy in the comprehensive detection of abnormalities on chest X-rays," *Scientific Reports*, vol. 14, 2024. [Online]. Available: <https://www.nature.com/articles/s41598-024-76608-2>
- [9] C. M. Jones *et al.*, "Assessment of the effect of a comprehensive chest radiograph deep learning model on radiologist reports and patient outcomes: A real-world observational study," *BMJ Open*, 2021. [Online]. Available: <https://pubmed.ncbi.nlm.nih.gov/34930738/>
- [10] J. Mongan *et al.*, "Artificial intelligence and machine learning for chest radiography," *Radiology*, vol. 307, no. 3, 2023. [Online]. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC10128969/>
- [11] E. Cheung *et al.*, "U'AI' testing: User interface and usability testing of a chest X-ray AI tool in a simulated real-world workflow," *Canadian Association of Radiologists Journal*, 2022. [Online]. Available: <https://journals.sagepub.com/doi/10.1177/08465371221131200>
- [12] A. J. DeGrave, J. D. Janizek, and S.-I. Lee, "AI for radiographic COVID-19 detection selects shortcuts over signal," *Nature Machine Intelligence*, vol. 3, pp. 610–619, 2021. [Online]. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC10047562/>
- [13] M. D. Danu, G. Marica, S. K. Karn, B. Georgescu, A. Mansoor, F. Ghesu, L. M. Itu, C. Suci, S. Grbic, O. Farri, and D. Comaniciu, "Generation of radiology findings in chest X-ray by leveraging collaborative knowledge," *Procedia Computer Science (ITQM 2023)*, 2023. [Online]. Available: <https://arxiv.org/abs/2306.10448>
- [14] C. Li *et al.*, "LLaVA-Med: Training a large language-and-vision assistant for biomedicine in one day," *Advances in Neural Information Processing Systems*, 2023. [Online]. Available: <https://arxiv.org/abs/2306.00890>
- [15] P. Lakhani and B. Sundaram, "Deep learning at chest radiography: Automated classification of pulmonary tuberculosis by using convolutional neural networks," *Radiology*, vol. 284, no. 2, pp. 574–582, 2017.
- [16] F. Pasa *et al.*, "Efficient deep network architectures for fast chest X-ray tuberculosis screening and visualization," *Scientific Reports*, vol. 9, 2019.
- [17] WHO Kobe Centre, "Determining the local calibration of computer-assisted detection (CAD) thresholds and other parameters." [Online]. Available: <https://wkc.who.int/resources/publications/i/item/determining-the-local-calibration-of-computer-assisted-detection-cad-thresholds-and-other-parameters>

- [18] A. Marquez *et al.*, "Performance of chest X-ray with computer-aided detection as a triage test for pulmonary TB in the Philippines," *BMC Global and Public Health*, 2025. [Online]. Available: <https://bmcbglobalpublichealth.biomedcentral.com/articles/10.1186/s44263-025-00198-y>
- [19] Qure.ai, "qXR documentation." [Online]. Available: <https://documentation.quire.ai/>
- [20] Delft Imaging, "CAD4TB solution." [Online]. Available: <https://delft.care/cad4tb-solution/>
- [21] Lunit, "INSIGHT CXR." [Online]. Available: <https://www.lunit.io/en/products/cxr>
- [22] Annalise.ai, "Enterprise CXR." [Online]. Available: <https://www.annalise.ai/CXR>
- [23] Siemens Healthineers, "AI-Rad Companion Chest X-Ray." [Online]. Available: <https://academy.siemens-healthineers.com/>
- [24] M. Nasser *et al.*, "Chest X-ray screening for pulmonary tuberculosis using photographed analog radiographs," *PLOS Digital Health*, 2023. [Online]. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC12955127/>
- [25] H. Chen, C. Gomez, C.-M. Huang, and M. Unberath, "Explainable medical imaging AI needs human-centered design: guidelines and evidence from a systematic review," *npj Digital Medicine*, vol. 5, no. 156, 2022. [Online]. Available: <https://www.nature.com/articles/s41746-022-00699-2>
- [26] K. Borys, Y. A. Schmitt, M. Nauta, C. Seifert, N. Krämer, C. M. Friedrich, and F. Nensa, "Explainable AI in medical imaging: An overview for clinical practitioners – Beyond saliency-based XAI approaches," *European Journal of Radiology*, vol. 162, art. 110786, 2023. [Online]. Available: <https://doi.org/10.1016/j.ejrad.2023.110786>
- [27] S. Lee, J. Youn, H. Kim, M. Kim, and S. H. Yoon, "CXR-LLaVA: a multimodal large language model for interpreting chest X-ray images," *European Radiology*, vol. 35, 2025. [Online]. Available: <https://link.springer.com/article/10.1007/s00330-024-11339-6>
- [28] I. Hartsock and G. Rasool, "Vision-language models for medical report generation and visual question answering: a review," *Frontiers in Artificial Intelligence*, vol. 7, art. 1430984, 2024. [Online]. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC11611889/>
- [29] U.S. Food and Drug Administration, "Methods and tools for effective postmarket monitoring of artificial intelligence (AI)-enabled medical devices," 2024. [Online]. Available: <https://www.fda.gov/medical-devices/medical-device-regulatory-science-research-programs-conducted-osel/methods-and-tools-effective-postmarket-monitoring-artificial-intelligence-ai-enabled-medical-devices>
- [30] J. Rudolph *et al.*, "Post-deployment performance of a deep learning algorithm for normal and abnormal chest X-ray classification: A study at visa screening centers in the United Arab Emirates," *PLOS ONE*, 2024. [Online]. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC11539241/>
- [31] A. S. Tejani, H. Elhalawani, L. Moy, M. Kohli, and C. E. Kahn Jr, "Artificial intelligence and radiology education," *Radiology: Artificial Intelligence*, vol. 5, no. 1, e220084, 2023. [Online]. Available: <https://pubs.rsna.org/doi/full/10.1148/ryai.220084>
- [32] M. A. Mazurowski, "Artificial intelligence for precision education in radiology," *British Journal of Radiology*, vol. 92, no. 1103, 2019. [Online]. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC6849670/>
- [33] S. Vijayan *et al.*, "Implementing a chest X-ray artificial intelligence tool to enhance tuberculosis screening in India: Lessons learned," *PLOS Digital Health*, vol. 2, no. 12, e0000404, 2023. [Online]. Available: <https://journals.plos.org/digitalhealth/article?id=10.1371/journal.pdig.0000404>
- [34] A. L. Innes *et al.*, "Computer-aided detection for chest radiography to improve the quality of tuberculosis diagnosis in Vietnam's district health facilities: An implementation study," *Tropical Medicine and Infectious Disease*, vol. 8, no. 11, art. 488, 2023. [Online]. Available: <https://doi.org/10.3390/tropicalmed8110488>
- [35] J. Creswell *et al.*, "Early user perspectives on using computer-aided detection software for interpreting chest X-ray images to enhance access and quality of care for persons with tuberculosis," *BMC Global and Public Health*, vol. 1, no. 30, 2023. [Online]. Available: <https://bmcbglobalpublichealth.biomedcentral.com/articles/10.1186/s44263-023-00033-2>
- [36] A. J. Scott *et al.*, "Diagnostic accuracy of computer-aided detection during active case finding for pulmonary tuberculosis in Africa: A systematic review and meta-analysis," *Open Forum Infectious Diseases*, vol. 11, no. 2, ofae020, 2024. [Online]. Available: <https://doi.org/10.1093/ofid/ofae020>
- [37] F. A. Khan *et al.*, "A systematic review and meta-analysis of artificial intelligence software for tuberculosis diagnosis using chest X-ray imaging," 2025. [Online]. Available: <https://pubmed.ncbi.nlm.nih.gov/40529749/>
- [38] "Domain shift analysis in chest radiographs classification in a Veterans Healthcare Administration population," *Journal of Imaging Informatics in Medicine*, 2025. [Online]. Available: <https://link.springer.com/article/10.1007/s10278-025-01494-7>
- [39] I. F. Ihongbe, S. Fouad, T. F. Mahmoud, A. Rajasekaran, and A. Bhatia, "Evaluating explainable artificial intelligence (XAI) techniques in chest radiology imaging through a human-centered lens," *PLOS ONE*, vol. 19, no. 10, e0308758, 2024. [Online]. Available: <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0308758>
- [40] S. Gaube *et al.*, "Care to explain? AI explanation types differentially impact chest radiograph diagnostic performance and physician trust in AI," *Radiology*, vol. 313, no. 1, e233261, 2024. [Online]. Available: <https://pubs.rsna.org/doi/10.1148/radiol.233261>
- [41] S. Tonekaboni, S. Joshi, M. D. McCradden, and A. Goldenberg, "What clinicians want: Contextualizing explainable machine learning for clinical end use," in *Proc. Machine Learning for Healthcare (MLHC)*, 2019. [Online]. Available: <https://arxiv.org/abs/1905.05134>
- [42] C. J. Cai, S. Winter, D. Steiner, L. Wilcox, and M. Terry, "Hello AI: Uncovering the onboarding needs of medical practitioners for human-AI collaborative decision-making," *Proc. ACM Human-Computer Interaction (CSCW)*, vol. 3, art. 104, 2019. [Online]. Available: <https://dl.acm.org/doi/10.1145/3359206>
- [43] D. Hua *et al.*, "Towards human-AI collaboration in radiology: A multidimensional evaluation of the acceptability of AI for chest radiograph analysis in supporting pulmonary tuberculosis diagnosis," *JAMIA Open*, vol. 8, no. 1, ooae151, 2025. [Online]. Available: <https://academic.oup.com/jamiaopen/article/8/1/ooae151/8002370>
- [44] G. Chassagnon *et al.*, "Learning from the machine: AI assistance is not an effective learning tool for resident education in chest X-ray interpretation," *European Radiology*, 2023. [Online]. Available: <https://pubmed.ncbi.nlm.nih.gov/37572190/>
- [45] Y. Liu, Y.-H. Wu, Y. Ban, H. Wang, and M.-M. Cheng, "Rethinking computer-aided tuberculosis diagnosis," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 2646–2655. [Online]. Available: <https://mmcheng.net/tb/>
- [46] Y. Liu, Y.-H. Wu, S.-C. Zhang, L. Liu, M. Wu, and M.-M. Cheng, "Revisiting computer-aided tuberculosis diagnosis," *IEEE Trans. Pattern Analysis and Machine Intelligence*, 2023. [Online]. Available: <https://arxiv.org/abs/2307.02848>
- [47] F. Yang, P.-X. Lu, M. Deng *et al.*, "Annotations of lung abnormalities in the Shenzhen chest X-ray dataset for computer-aided screening of pulmonary diseases," *Data*, vol. 7, no. 7, art. 95, 2022. [Online]. Available: <https://www.mdpi.com/2306-5729/7/7/95>
- [48] J. P. Cohen, J. D. Viviano, P. Bertin *et al.*, "TorchXRyVision: A library of chest X-ray datasets and models," in *Proc. Medical Imaging with Deep Learning (MIDL)*, 2022. [Online]. Available: <https://arxiv.org/abs/2111.00595>
- [49] A. Bustos, A. Pertusa, J.-M. Salinas, and M. de la Iglesia-Vayá, "PadChest: A large chest X-ray image dataset with multi-label annotated reports," *Medical Image Analysis*, vol. 66, art. 101797, 2020. [Online]. Available: <https://arxiv.org/abs/1901.07441>
- [50] S. G. Finlayson *et al.*, "The clinician and dataset shift in artificial intelligence," *New England Journal of Medicine*, vol. 385, pp. 283–286, 2021. [Online]. Available: <https://pubmed.ncbi.nlm.nih.gov/34260843/>
- [51] J. Feng, R. V. Phillips, I. Malenica *et al.*, "Clinical artificial intelligence quality improvement: Towards continual monitoring and updating of AI algorithms in healthcare," *npj Digital Medicine*, vol. 5, art. 66, 2022. [Online]. Available: <https://www.nature.com/articles/s41746-022-00611-y>
- [52] J. Merkow *et al.*, "Empirical data drift detection experiments on real-world medical imaging data," *Nature Commu-*

nications, vol. 15, art. 1689, 2024. [Online]. Available:
<https://www.nature.com/articles/s41467-024-46142-w>

- [53] A. Ghosh *et al.*, “Scalable drift monitoring in medical imaging AI,” *arXiv preprint arXiv:2410.13174*, 2024. [Online]. Available:
<https://arxiv.org/abs/2410.13174>